

Yasha Duncan-Wilson

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A highly self-motivated and results focused individual. Displays strong analytical and problem solving skills, with meticulous attention to detail and effective decision making; key parts of past roles have involved understanding the detail of software used by end-users and how to solve problems they may encounter while using it. Equally comfortable working as part of a team or independently, can be relied upon to meet tight deadlines and collaborate with others when working in less familiar areas. Committed to skills improvement and keen to increase technical capability, with the ability to learn new concepts quickly.

Skills

Programming: Python, SQLite, HTML, CSS, JS, React

Technology: Docker, macOS, Linux, Windows, Git, TDD, CI/CD

Languages: Fluent English with excellent spelling and grammar

Experience

Software Developer Freelance – Edinburgh 2020 – Present

- Worked as a solo developer, taking ownership of client experience and whole product development.
- Developed a variety of products in Python (with FastAPI and SQLite) and React (with HTML/CSS).
- Provided fast iteration during development phase and high attention to detail when tailoring UI/UX to match client branding.

Director Edinburgh Hacklab – Edinburgh 2025 – Present

- Volunteering to maintain a welcoming, friendly and safe community space for members and guests.
- Leading efforts to improve documentation and training for existing tooling.
- Managing the demands of an opinionated and vocal membership during a transitional period.

Team Lead Heriot-Watt University CanSat Society – Edinburgh 2024 – 2025

Mechanical Engineer Heriot-Watt University CanSat Society – Edinburgh 2023 – 2024

- Selected and led a team to compete in the American Aeronautical Society CanSat Competition, managing budget, deadlines and team dynamics as well as engineering requirements.
- Performed analyses of competing solutions to complex problems, ultimately choosing designs which were fit for purpose while meeting mass and material restrictions.
- Adapted aerospace industry design patterns and analytic tools such as FMEA and RCA for use by a small team.

Summer Intern Heriot-Watt University GRID facility – Edinburgh 2024

- Designed and implemented solution for reliable automatic 3D printer bed clearing with no intervention.
- Softwareless print bed clearing uses existing printer electronics, requiring only minor modifications
- Improved throughput by reducing the need for lab technicians to clear prints and reset printers.

Education

Heriot-Watt University – MEng in Aerospace Engineering 2023-2029

Projects

Smartshelf <https://github.com/iykdw/smartshelf>

- A Python and FastAPI webapp with SQLite for determining optimal book positions on home bookshelves
- Calculates optimal book placement and tracks borrowing for multiple users using Remote-User header forwarding
- Project is containerised using Docker for ease of deployment
- Deployed on home server via `docker compose` using Authelia for authentication and Caddy as a reverse proxy

Personal Server

- A server which hosts media, webapps (including Smartshelf), and backups off-site
- Runs Docker on Debian and software RAID with Authelia and LLDAP for authentication

Stickerpacker <https://github.com/iykdw/stickerpacking>

- A Python app which efficiently places stickers in a bounded area and generate cut paths
- Uses `matplotlib`, `imagemagick`, `pydantic` to determine image boundaries for exporting to Cameo CNC cutter
- Deployed in Edinburgh Hacklab with regular use

Futures Scanner <https://wmcscanner-87do9.web.app/>

- A React app developed in close collaboration with a futures consultancy as a potential product
- Closely matches existing client documents and meets design guidelines

Ender 3 Klipper Conversion <https://github.com/iykdw/klipper-ender-3>

- Improved poor print success rates: built and fitted an ABL probe to improve first-layer reliability
- Upgraded firmware: wrote and tuned Klipper config to improve print outcomes and user experience